DFT and QFT: Same Frequency Story, Different Language

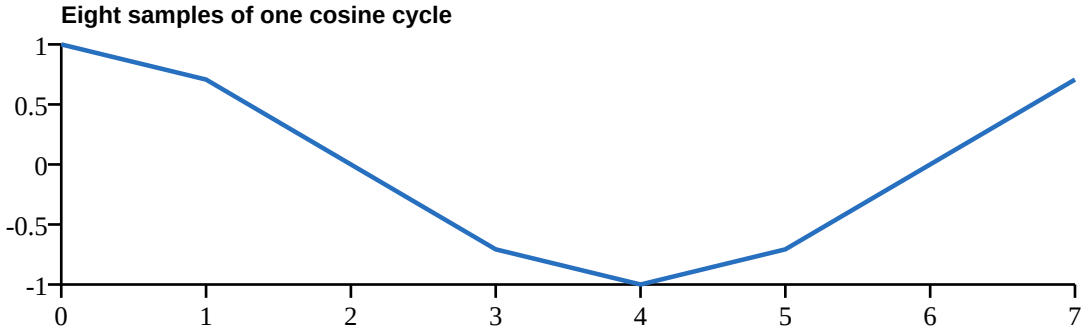
A worked classroom demonstration using eight samples of one cosine wave

Learning goal. Show that DFT and QFT share the same Fourier mathematics, but return different kinds of outputs: classical complex coefficients versus amplitudes of a quantum state.

1. Start with one simple signal

Choose f[n] = cos(2πn/8), for n = 0, 1, …, 7. It completes exactly one cycle over these eight samples.

n	Angle	f[n] exact	Approx.
0	2π(0)/8	1	1.0000
1	2π(1)/8	1/√2	0.7071
2	2π(2)/8	0	0.0000
3	2π(3)/8	–1/√2	-0.7071
4	2π(4)/8	–1	-1.0000
5	2π(5)/8	–1/√2	-0.7071
6	2π(6)/8	0	-0.0000
7	2π(7)/8	1/√2	0.7071



Analogy: imagine taking eight snapshots of a repeating sound. The Fourier transform compares those values with repeating reference patterns to find which frequencies match.

2. Classical DFT: calculate the frequency bins

Let ω = e^(2πi/8) = e^(iπ/4) = (1+i)/√2. We use the positive-exponent, unitary-normalized convention:

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X[k] = (1/√8) Σ(n=0 to 7) f[n] ω^(kn), k = 0,…,7
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Other textbooks/software may use a negative exponent or put normalization elsewhere. Frequency-bin locations and magnitudes are unchanged.

Useful powers of ω

Power	Value	Power	Value
ω■	1	ω■	–1
ω¹	(1+i)/√2	ω■	(–1–i)/√2
ω²	i	ω■	–i
ω³	(–1+i)/√2	ω■	(1–i)/√2

Manual calculation A: DC bin k = 0

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X[0] = (1/√8)(1 + a + 0 – a – 1 – a + 0 + a) = 0, where a = 1/√2.
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The positive and negative samples cancel: there is no constant offset.

Manual calculation B: bin k = 1

$X[1] = (1/\sqrt{8})[1 + a\omega - a\omega^3 - \omega\blacksquare - a\omega\blacksquare + a\omega\blacksquare]$.

Substitute: $a\omega=(1+i)/2$; $-a\omega^3=(1-i)/2$; $-\omega\blacksquare=+1$; $-a\omega\blacksquare=(1+i)/2$; $a\omega\blacksquare=(1-i)/2$. The bracket sums to 4.

Therefore $X[1] = 4/\sqrt{8} = \sqrt{2} \approx 1.4142$. Similarly $X[7] = \sqrt{2}$; all other bins are zero.

Normalized DFT result: $X = [0, \sqrt{2}, 0, 0, 0, 0, 0, \sqrt{2}]\blacksquare$. The peaks at k=1 and k=7 are the positive- and negative-frequency components of the same real cosine.

3. The same information in a 3-qubit QFT

Three qubits have $2^3 = 8$ basis states, $|000\rangle$ through $|111\rangle$. Eight samples fit naturally; nine samples do not fit directly into a 3-qubit register.

The classical sample vector has squared norm $\Sigma|f[n]|^2 = 4$, so normalize it to make a valid quantum state:

$$|\psi\rangle = (1/2)\ \Sigma_{n=0 \text{ to } 7}\ f[n]|n\rangle$$

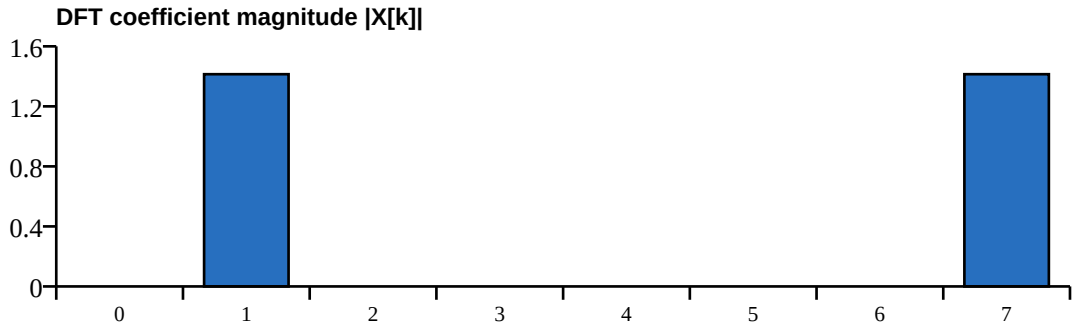
The QFT maps $|n\rangle$ to $(1/\sqrt{8})\ \Sigma_{k=0 \text{ to } 7}\ e^{(2\pi ikn/8)}|k\rangle$. Thus the output amplitude is:

$$A[k] = (1/\sqrt{8})\ \Sigma\ [f[n]/2]\ \omega^{(kn)} = X[k]/2$$

So the output state is: $|\psi_{out}\rangle = (|001\rangle + |111\rangle)/\sqrt{2}$.

Why do the numbers differ?

The DFT gives coefficients $\sqrt{2}$ in bins 1 and 7. A quantum state must have total probability 1, so the corresponding QFT amplitudes are $1/\sqrt{2}$. Their squared magnitudes are probabilities: $1/2 + 1/2 = 1$.



On repeated quantum measurements, ideally 50% of outcomes are 001 and 50% are 111. A measurement reveals a bin label, not the full complex coefficient; phase information generally requires additional measurement strategies.

4. What does frequency mean in real units?

If the sample interval is Δt, the sample rate is $f_s = 1/\Delta t$. The frequency spacing is $f_s/8$. Bins 1 and 7 represent $+f_s/8$ and $-f_s/8$ in a two-sided spectrum.

Example: $\Delta t = 0.01$ s gives $f_s = 100$ Hz, so the cosine frequency is $100/8 = 12.5$ Hz. The two-sided spectrum shows +12.5 Hz and −12.5 Hz.

5. Student-friendly analogy

DFT: like checking a sound against eight tuning forks. Each complex coefficient tells how strongly a reference pattern matches, including phase.

QFT: applies the same Fourier pattern to the amplitudes of a quantum state. Measurement samples the frequency-bin outcomes; it does not print every coefficient.

Same: Fourier kernel, bins, and underlying mathematics. **Different:** DFT processes a classical vector and returns all coefficients; QFT transforms a normalized quantum state and measurements give sampled outcomes.

Whiteboard summary

$$f[n] = \cos(2\pi n/8) \rightarrow \text{DFT peaks at } k = 1, 7 \rightarrow \text{QFT state } (|001\rangle + |111\rangle)/\sqrt{2}.$$

Convention note: this handout uses the positive exponent to match the QFT formula. A common classical DFT convention uses a negative exponent; it conjugates the coefficients. For this real cosine, the peak locations and magnitudes remain the same.